FLYLORA

FlyLoRA

Implicit Rank-Wise Mixture-of-Experts for Efficient Adaptation | NeurIPS 2025

Inspired by the fly olfactory circuit, freeze a sparse random LoRA down-projection and use its top-k activations as an implicit rank-wise router, reducing within-task and cross-task interference.

LoRA · implicit MoE · model merging

1 Problem

RANK INTERFERENCE

- LoRA needs higher rank for complex tasks, but correlated rank updates waste capacity and can destabilize training.
- MoE-LoRA separates experts but pays for a trainable router and still mixes ranks within each expert.
- Merging task-specific adapters introduces destructive inter-task interference.

2 Bio-inspired design

- The fly olfactory circuit expands inputs through sparse random projections, then keeps only strong activations.
- FlyLoRA freezes sparse A, computes Ax, and activates the k rank-1 columns of B linked to largest magnitudes.
- A loss-free bias balances usage; A serves simultaneously as down-projection and router.

3 Theory

DECOUPLING BY SPARSITY

- Sparse random A approximately preserves pairwise distances, so similar inputs route consistently.
- `off-diagonal gradient covariance shrinks by about k^2/r^2`
- Independent task projections are nearly orthogonal, making merged adapter updates approximately non-interfering.

4 Efficiency

- Total rank r=32 offers a broad subspace while only k=8 rank experts are active per token.
- Freezing A removes optimizer and activation-gradient cost; eliminating Wg removes router parameters.
- Top-25% projected dimensions empirically retain over 80% of activation energy.

5 Evidence

SINGLE + MULTI-TASK GAINS

- On Llama-3.1-8B, 0.13% active trainable parameters beat LoRA and Split-LoRA on all four domains.
- Results: MMLU 40.88, ScienceQA 94.15, GSM8K 58.76 and HumanEval pass@1 36.88.
- After simple multi-task merging it retains substantially more accuracy; frozen A is crucial.

6 Meaning & limits

GEOMETRY AS ROUTING

- Fixed random geometry can replace a learned router while creating useful task-separated subspaces.
- Evidence focuses on four benchmarks and two 7–8B backbones; scaling needs wider validation.
- Random projections reduce interference but cannot guarantee semantic specialization or lossless merging.